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Lea Frost

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EDUCATION

University of South Carolina Honors College – Columbia, SC
B.S. Computer Science, B.S. Data Science

Expected May 2027
GPA: 3.93

LANGUAGES AND TECHNOLOGIES

| Python | R | C++ | SQL | Java | Jupyter | Tableau | Hadoop | Pandas | Plotly | Matplotlib | PyTorch | Unix/Linux | VSCode | Git |

COURSEWORK

Multivar. Statistics & Data Mining | Database System Design | Artificial Intelligence | Big Data Analytics | Visualization Tools |
Linear Algebra | Vector Calculus | Discrete Math | Data Structures & Algorithms | Adv. Prog. Techniques | Software Eng. |

RELEVANT EXPERIENCE

University of South Carolina – *Undergraduate Research Assistant* Aug 2025 – Present
Funded by Office of Naval Research (ONR) | Title: *Advanced Perception for Autonomous Platforms in the Littorals*

- Contributing to a project that aims to improve object detection technologies in difficult weather conditions.
- Trains state-of-the-art machine learning models by generating training data utilizing image labeling software.
- Collaborates with other researchers to improve results and trouble-shoot image labeling problems.

University of South Carolina NSF REU – *Mathematical Foundations of Data Analysis*

June 2025 – July 2025

- Contributed to a project studying colon cancer reoccurrence among ICU patients using biomarker time-series data.
- Studied various time-series to image encoding methods and trained a Visual Transformer using generated images.
- Collaborated with graduate researchers to improve classification results with a very small and incomplete dataset.

June 2024 – July 2024

- Developed a PyTorch deep learning model for proving various mathematical graph theory conjectures.
- Held weekly presentations with REU participants explaining graph theory and our deep learning model.
- Contributed to a research paper written by our research group detailing our models and their results.

University of South Carolina Dep. of IT – *Enterprise Data Networks Student Team Member* Apr 2025 – May 2025

- Assisted the Department of Information Technology with organizing and tracking decommissioned assets.
- Performed manual factory resets of assets, primarily network switches, using PuTTY serial console.
- Optimized the factory resetting process by writing a Python script utilizing the pySerial library.

PROJECTS

Language Learning Application Aug 2024 – Dec 2024

- Java/JavaFX application imitating interactive language learning applications like Duolingo.
- Created both the backend and GUI from scratch with a team of developers in a scrum framework.
- Supports user account creation and management and securely saves user data in JSON file format.
- Applied object-oriented programming principles to create modular, maintainable, and scalable code.

Business Chatbot Feb 2024– Apr 2024

- Python-based hardcoded chatbot that answers questions related to a business based on web-scraped 10-k Form.
- Processes 10-k format, handles any user input, and report statistics on the interaction when program is terminated
- Chosen by the professor as the exemplary project of the class, displayed on the class website.

ACTIVITIES

Tau Beta Pi, Beta Chapter – *Treasurer* May 2025 – Present

- Manages chapter dues and expenditures and completes financial forms adhering to national deadlines.

HONORS/AWARDS

- Chubb Foundation Scholarship Recipient Fall 2024 – Present
- USC President's Honor List Fall 2023 – Fall 2024
- USC Dean's List Spring 2025
- Tau Beta Pi – Engineering Honor Society Spring 2024 – Present